CALVIN GACHERU MWANGI

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Portfolio: https://calvin-gacheru.github.io/MyPortfolio/

PROFESSIONAL PROFILE

Data Scientist, AI Engineer, and Full-Stack Developer specializing in the complete software lifecycle. Focused on transitioning machine learning models and AI systems from local environments to production-ready deployments. Expertise in building robust Retrieval-Augmented Generation (RAG) architectures, training predictive algorithms, and engineering enterprise ETL pipelines.

EDUCATION

BSc. Data Science & Analytics

United States International University — Africa (USIU-Africa), Nairobi, Kenya

Expected Graduation: 2027

TECHNICAL PROFICIENCIES

- **Programming Languages:** Python, SQL, JavaScript, HTML5/CSS3, Bash, R
- **Artificial Intelligence & Machine Learning:** Large Language Models (Llama 3.1, Groq), LangChain, LangGraph, Agentic Orchestration, RAG Systems, Scikit-learn (Random Forest, XGBoost), NLP, Time Series Analysis, Predictive Analytics, Explainable AI (SHAP)
- **Data Engineering & Infrastructure:** Vector Databases (Qdrant), Docker, Kubernetes, Git/GitHub Actions (CI/CD), PostgreSQL, SQLite, Redis, Pandas, NumPy, Astral UV
- **Web Frameworks & OS:** FastAPI, Flask, Django, Streamlit, Linux/Fedora

PROFESSIONAL EXPERIENCE

BikBat Insurance Agency | Freelance Web Developer

2026

- Engineered, developed, and deployed a complete corporate web presence for a commercial insurance client.
- Managed the full deployment lifecycle, including domain registration, DNS configuration, and cloud hosting infrastructure.

ProdigyInfoTech | Data Science Virtual Intern

2025

- Executed four comprehensive data science projects utilizing real-world datasets.
- Performed exploratory data analysis, built predictive models, conducted customer churn analytics, and implemented NLP-based sentiment analysis with fully reproducible workflows.

KEY ENGINEERING PROJECTS

MySaidi - AI Productivity Assistant

- Architected a full-stack Progressive Web App (PWA) operating on a decoupled browser-to-REST API architecture.
- Integrated a LangGraph-powered AI agent using Groq LLMs to manage user schedules, tasks, and habits via natural language intent parsing.
- Built the backend with FastAPI and PostgreSQL, and designed a mobile-optimized vanilla JavaScript frontend secured with JWT authentication.

Uber Price Prediction & Production MLOps Pipeline

- Engineered an end-to-end machine learning pipeline predicting ride prices using an optimized Random Forest Regressor (Cross-Validation Average R^2 Score: 0.9854).
- Prevented data leakage by bundling transformations into Scikit-Learn pipelines.
- Containerized the ecosystem using Docker and orchestrated automated CI/CD workflows via GitHub Actions.
- Served the inference model through a FastAPI REST backend with Pydantic validation, accessed via an interactive Streamlit UI.

Behavioural Explainer - RAG System

- Developed a production-grade Python Retrieval-Augmented Generation (RAG) system for psychological framework analysis.
- Built a document processing pipeline utilizing PyMuPDF for text extraction and chunking, storing vector embeddings in a Qdrant database.
- Executed local LLM analysis via Ollama (Llama 3.1) to synthesize structural breakdowns of human interaction scenarios without offering generic advice.

- Packaged the tool with a custom Typer Command Line Interface and Docker.

Maternal Risk Detection System

- Trained an XGBoost and Random Forest ensemble model to classify maternal health risk tiers from patient vitals and historical data.
- Implemented real-time outlier detection and SHAP explainability to map feature influence on clinical predictions.
- Developed a feedback loop automating SMS appointment reminders and wellness tips using the Africa's Talking API and APScheduler.

Interactive Web Platforms (Arcade Kenya & Valen Time Love)

- Developed "Arcade Kenya," a highly interactive, animated platform utilizing HTML5 Canvas and CSS3 for an entertainment venue.
- Built "Valen Time Love," a customizable SaaS-style web application framework generating dynamic, shareable event proposals.

ETL Data Engineering

- Engineered custom ETL processes handling structured datasets for automated extraction, transformation, and loading, complete with validation and logging mechanisms.

CERTIFICATIONS

- **Building AI Agents using LangGraph** – Simplilearn
- **Data Science Virtual Internship** – ProdigyInfoTech